

WEEK 4 – EXPLORATORY DATA ANALYSIS (EDA)

OBJECTIVE:

The objective of this analysis is to explore the Shopify stock dataset using Python, Pandas, Matplotlib and Seaborn. The analysis focuses on stock price trends, trading volume, moving averages, daily price changes, daily returns and volatility.

ANALYSIS:

Analysis	Description
Stock Price Analysis	Open, High, Low and Close prices were visualized to understand Shopify stock price movement over time.
Trading Volume Analysis	Trading volume was plotted to observe changes in market activity over time.
Moving Average Analysis	20-Day and 50-Day Simple Moving Averages were calculated and compared with the closing price.
Daily Price Change	Trading volume was compared with Daily Price Change using a scatter plot.
Return Analysis	Mean, variance and standard deviation of Daily Return were calculated.
Return Distribution	Histogram and KDE plots were created to understand the distribution of Daily Return.
Volatility Analysis	The 90th percentile of absolute Daily Return was used to identify high-volatility observations.

KEY FINDINGS:

- Open, High, Low and Close prices show changes in Shopify stock prices over time.
- Trading volume varies across different periods.
- 20-Day and 50-Day moving averages help in observing price trends.
- Daily Return shows the percentage movement in stock price.
- Histogram and KDE plots show the distribution of Daily Returns.
- High-volatility observations were identified using the 90th percentile method.

MAJOR INSIGHTS:

- Shopify stock prices show noticeable movement over time.
- Trading volume and daily price change can be compared to understand market activity.
- Moving averages provide a smoother view of the closing-price trend.
- Daily Return statistics help understand the variation in stock performance.
- The volatility analysis identifies days with comparatively large price movements.

RECOMMENDATIONS:

- Further analyze the relationship between trading volume and Daily Return.
- Compare the 20-Day and 50-Day moving averages for different time periods.
- the identified high-volatility observations separately.
- Perform additional correlation and time-series analysis for deeper insights.

CONCLUSION:

The Week 4 EDA successfully analyzed the Shopify stock dataset using different statistical and visualization techniques. The analysis provides insights into stock price trends, trading volume, moving averages, daily returns and volatility. These findings can be used for further analysis of Shopify stock performance.